

Department For System Engineering And Cyber Security

Instructions for the use of internal cloud infrastructure

Algebra Bernays University

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Introduction

Dear fellow students,

In front of you is an internal document of the Chair of Operating Systems of Algebra Bernays University, which explains the methodology and way of using the internal cloud infrastructure for the needs of the systems-oriented courses of Algebra Bernays University. The use of internal cloud infrastructure completely eliminates the problem of installing virtual machines on each individual computer, and opens up a number of other possibilities – for example, for midterm exams you will be able to practice directly within the cloud infrastructure.

Our idea is to provide you with a simpler approach to exercises as well as to practice for midterm exams, all in order to improve the quality of teaching and your degree of satisfaction with the study.

We hope that the use of cloud infrastructure will be a relief for you in passing the courses of the Chair of Operating Systems. Good luck!

Sincerely,
Vedran Dakić, mag.ing.el.
Head of the Operating Systems Department
Algebra Bernays University

1. Prerequisites

In order to connect to the cloud infrastructure, you need:

- a PC running Windows 7 or higher (it is also possible to use macOS, Linux, Android and iOS, by downloading an adequate version of the FortiClient VPN application);
- an installed FortiClient VPN client (you can download it from <https://www.fortinet.com/support/product-downloads#vpn>);
- if you do not want to use the Web Console (a tab in the browser), download VMware Remote Console from the following URLs:
 - Windows version – shorturl.at/kqWY7
 - Linux version – shorturl.at/gNRY0
 - macOS version – download from the App Store

In short, FortiClient VPN and a web browser (Chrome or Firefox) are enough for you to access it. VMRC is an optional component, with which certain functionalities work better.

The installation of all applications follows a next-next-next principle.

Using VPN on enterprise managed devices is not recommended.

2. FortiClient VPN Client Configuration

For the purpose of connecting to the environment, after installing the FortiClient VPN client, it is necessary to perform a configuration. The essential configuration data are as follows:

IPsec PSK: Pa\$\$w0rdPa\$\$w0rd123

User name: vuastudent

Password: St00dent123

IPsec LocalID: 4Students

The configuration is very simple, as will be explained through the following four images:

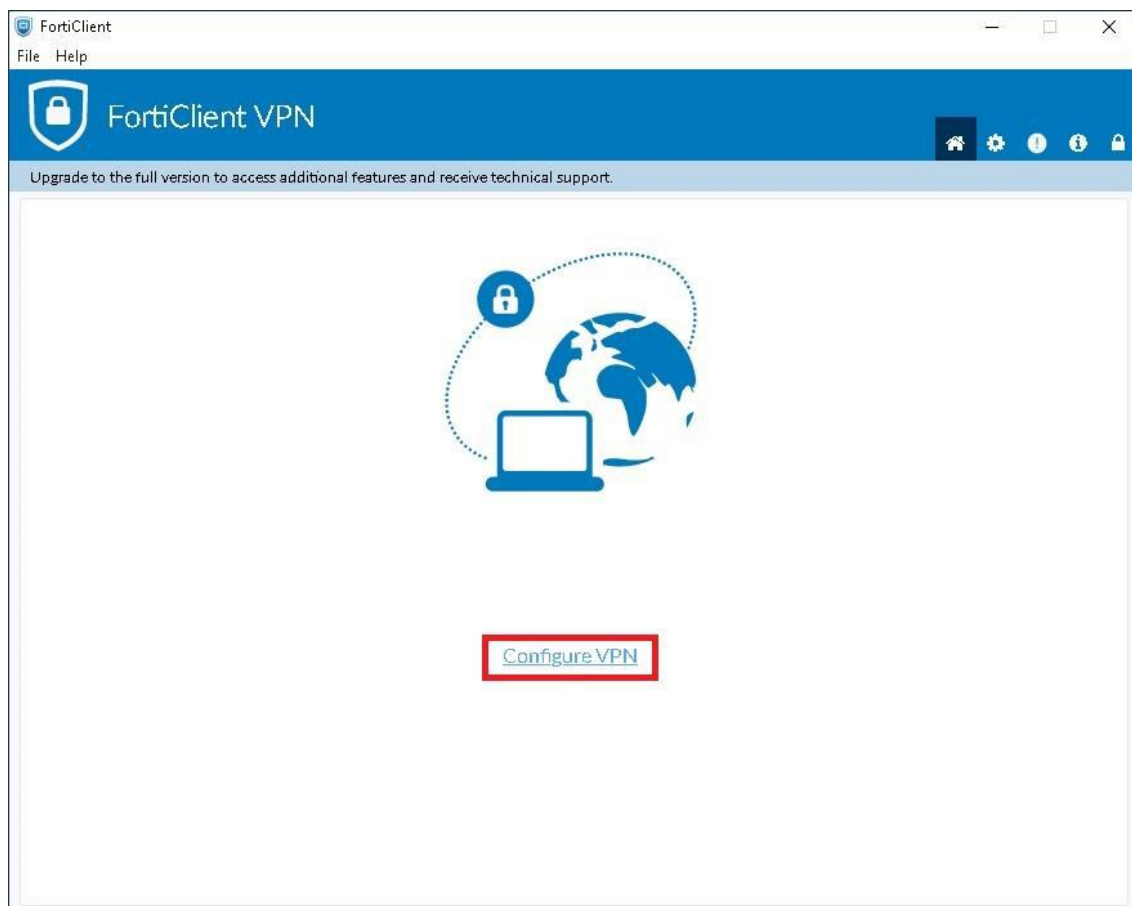


Figure 1 - Click on "Configure VPN".

After confirmation, it is necessary to select "Configure VPN", as shown in Figure 1. After that, it is necessary to select "IPsec VPN", as shown in Figure 2:

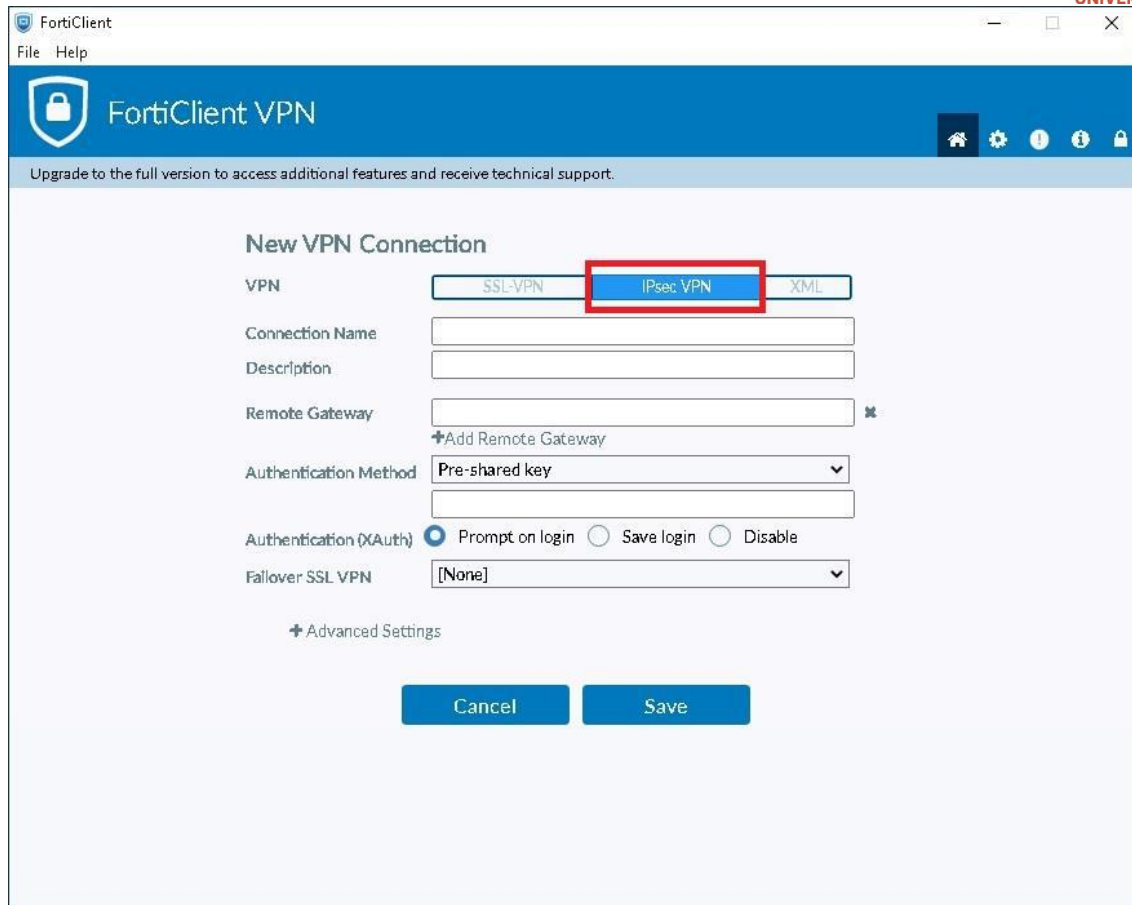
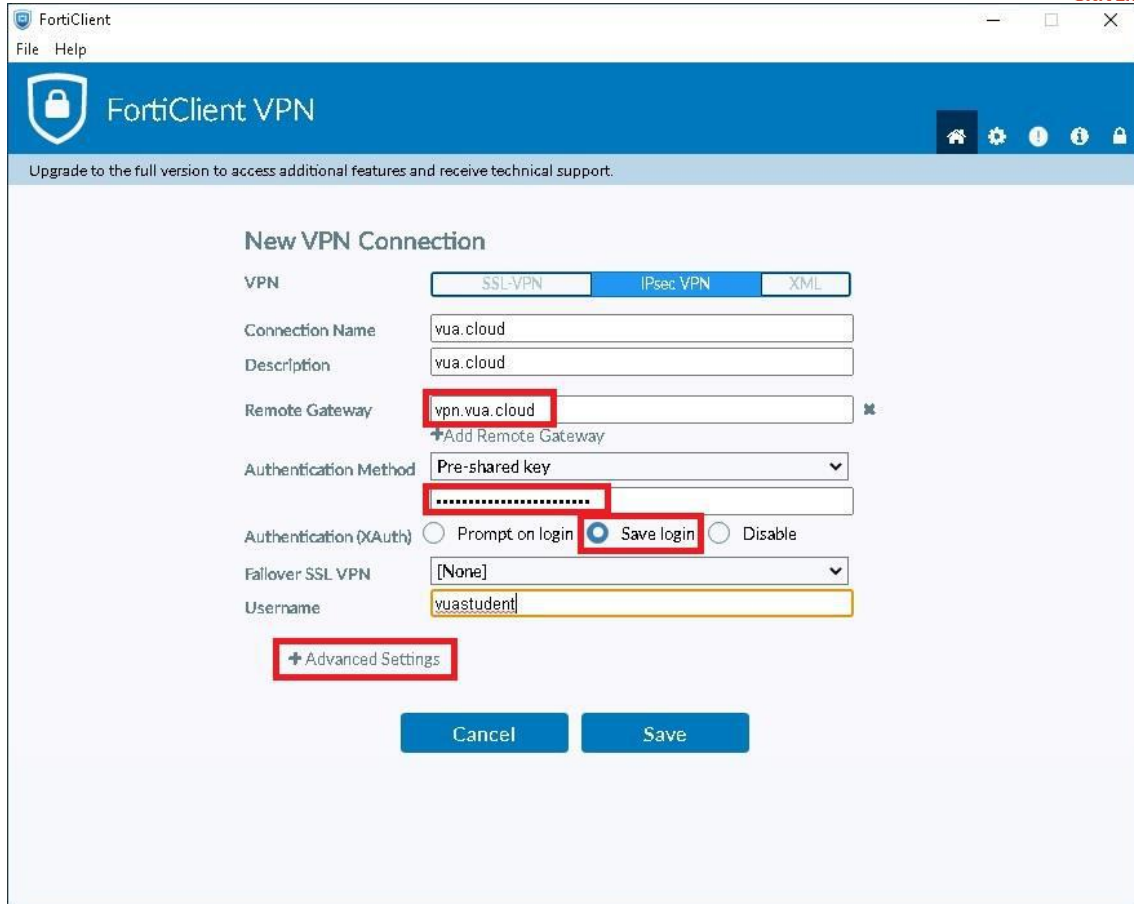


Figure 2 - Select the “IPsec VPN” option.

The next step is to set the basic parameters, as shown in Figure 3. The “Connection Name” and “Description” parameters are irrelevant, i.e. you can set them to any value. The “Remote Gateway” parameter must be set to `vpn.vua.cloud` and the “Authentication Method” parameter to “Pre-shared key”, the value of which must be `Pa$$w0rdPa$$w0rd123`:



FortiClient
File Help

FortiClient VPN

Upgrade to the full version to access additional features and receive technical support.

New VPN Connection

VPN: SSL-VPN IPsec VPN XML

Connection Name:

Description:

Remote Gateway: + Add Remote Gateway

Authentication Method: Pre-shared key ▼

Authentication (XAuth): ☐ Prompt on login ☒ Save login ☐ Disable

Fallover SSL VPN: [None] ▼

Username:

+ Advanced Settings

Cancel Save

Figure 3 - Parameter selection

In the next step we open additional options under “Advanced Settings”, so we click with the mouse on the “+” sign next to that option, and on the next screen we repeat the same in the “Phase 1” option:

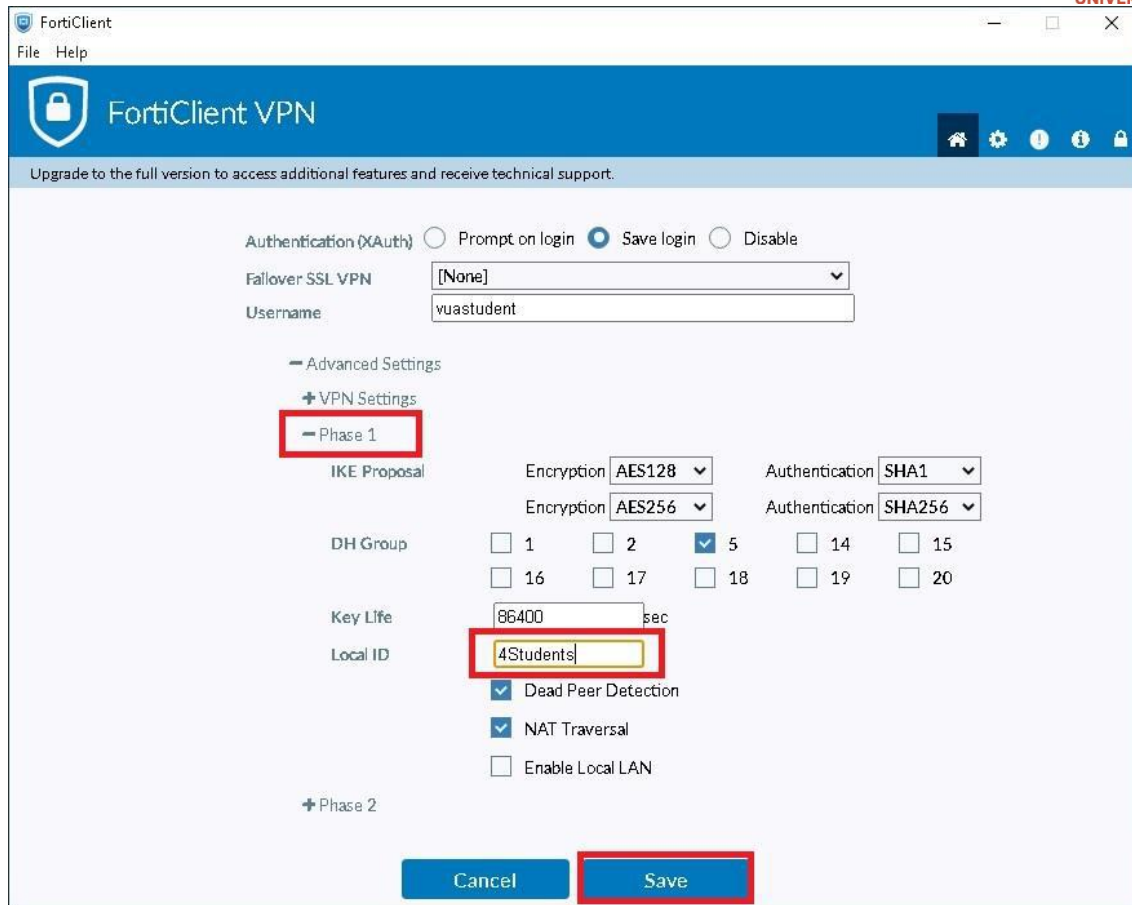


Figure 4 - Phase 1 - Local ID config

In the “Local ID” field we enter 4Students, after which we can press “Save”.
 In the next step we can start the VPN connection, and when asked, enter the password (St00dent123).
 After a successful connection, we can open the web browser and start using the cloud environment.

Using the cloud environment

2.1. Connecting to cloud infrastructure “from the outside”

Due to the fact that the complete cloud infrastructure is located within the campus of Algebra Bernays University, to connect to the cloud infrastructure “from home” we connect through a VPN client into the cloud environment, open the web browser and connect to <https://vcsa7.vua.cloud/ui>. You will receive a username and password for access during the first lab exercises of the first course in the third semester.

2.2. Connecting to cloud infrastructure during exercises

For the purpose of performing exercises, access to the vua.cloud infrastructure is enabled directly through a web browser. At the time of writing these instructions, the best results are achieved using the Chrome web browser. We run classroom computers on the Windows 10 operating system, and log in with a student user account and a student password. It is a non-mandatory roaming account profile that is made only for the needs of students who use vua.cloud for exercises.

In general, usernames for vua.cloud are your Algebra usernames without the @algebra.hr part. Example [student@algebra.hr](#) enters student as his username. Password is Pa\$\$w0rd.

Since there is a possibility that two students of the same first and last name may appear, there may also be deviations from this general rule.

When logging into the classroom computer, start your Chrome web browser and connect to the following address:

<https://vcsa7.vua.cloud/ui>

You will receive a login window in which you will enter your vua.cloud username and password, in the format username and password. When you confirm your data in the window and log in successfully, you will get GUI access to the cloud environment.

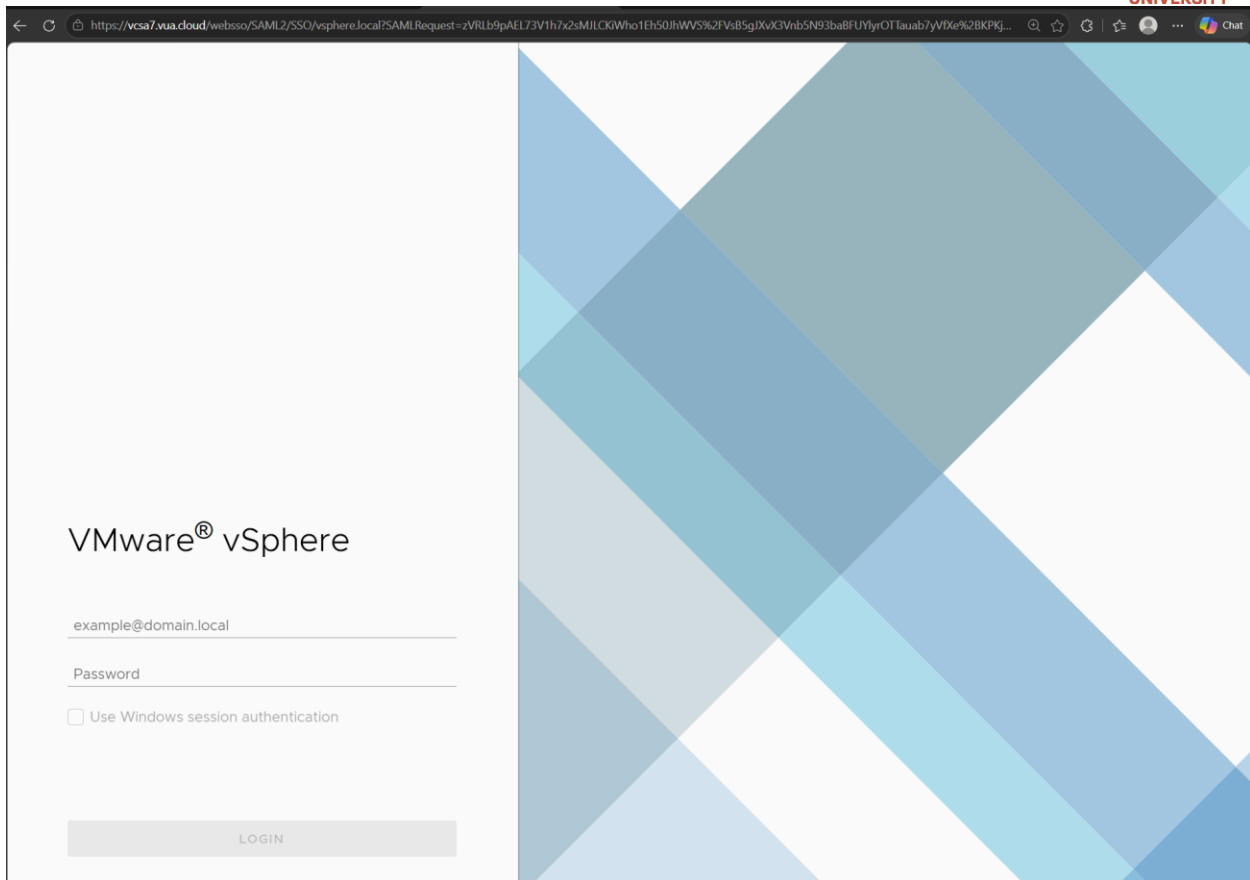


Figure 5 – vSphere login page

After that, you will switch to the “VMs and Templates” view so that you can access the folder with your virtual machines (e.g. Ilica, and sub-folder 2022). In this folder there is a folder with the year tag, the index mark of the student and the username of the student. Use the machines that are in that folder for practice and exams. Assistants will explain all the further steps – opening consoles to virtual machines and all other details.

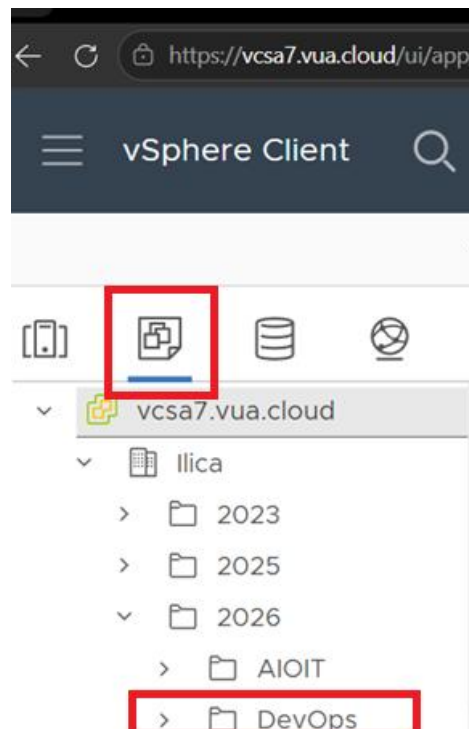


Figure 6 - vSphere view

2.3. Rules of conduct

Given the complexity of the infrastructure and the number of students using the cloud infrastructure, students are kindly asked to:

- not “share” your virtual machines with other students, which includes usernames and passwords for accessing the cloud infrastructure;
- not intentionally try to “crash” virtual machines, because the goal of the cloud infrastructure is to reduce students’ dependence on physical and virtual machines, so it makes no sense for us to contribute to additional problems, as well as to the number of disciplinary applications;
- when you are done with the exercise / preparation for the exam / practice / making of the project, turn off the machines. We expect machines that are not in use for more than 1 hour to be shut down, otherwise we start with a warning.
- do not bypass enforced configuration

Students are in charge of “administering” their own virtual machines. Translated, if you are asked to prepare virtual machines for the purposes of midterm exams (e.g. raising a domain and joining a domain), you are obliged to do so. Also, in such infrastructure preparation tasks, it is required to make snapshots of virtual machines with a specific name, which includes both uppercase and lowercase letters (e.g. “Domain” is not the same as “domain”). If you don’t make the required preparation, you are considered to be starting from a snapshot called “Start”, which is the default state at the beginning of the semester.

– on virtual machines students have the right to make one snapshot during the semester, and the exercises are adapted to such a scenario. In no scenario do we allow additional snapshots to be taken;

– the infrastructure is under constant surveillance by definition, because background applications themselves work that way. For all intentional attempts to cause problems, expect the application of all available Regulations of Algebra Bernays University.

Examples include but are not limited to:

- installation of a torrent client in virtual machines in the infrastructure;
- deliberately booting a DHCP server in one of the publicly available networks;
- intentional use of vua.cloud infrastructure for spamming, writing fake e-mails, phishing and/or any other form of security attacks;
- raising Minecraft or Discord servers or any similar services;
- any attempt to break into or demolish virtual infrastructure management machines or the machines of any other vua.cloud student.

The vua.cloud environment is constantly being upgraded, so the resource allocation policy can also change during the semester or academic year. Assistants and professors will give you all relevant information related to this topic. In any case, we reserve the right to shut down the environment at any time if there are (according to us) justified reasons – security, network, power outage, hardware problems, and the like.

Also, during breaks between semesters, as well as during breaks due to summer vacations, the environment will always be shut down. For example, in the break between the winter and summer semesters, the environment will be shut down for about a week, and in the break between summer and autumn deadlines at least two weeks.

Be sure to include this in your preparations for exams or submitting projects.

2.4. Access to the vua.cloud environment for learning purposes outside the framework of regular classes

Access to the vua.cloud environment for the purpose of exercising during the semester (therefore, outside the term of the exercises and not at the time when access is already enabled for the purpose of practicing for the exam, on some “special, additional virtual machines”) will be granted as necessary and possible. Count on the fact that in the same environment there are about a hundred students working with over 1000 virtual machines and that it is not always possible to meet all the requirements. Of course, as much as we can, we will accommodate you.

3. Questions, compliments, problem reporting

For all questions, compliments and reports of problems, please send an e-mail to the address vedran.dakic@algebra.hr. Feel free to report any steps that have not worked for you in the exercises. Write the following details in the mail:

1. Your full name, group, which course it is and which term if it concerns exercises; if not, a general description of how the problem occurred;
2. What exactly the exercise is (number and name of the exercise);
3. A step of the exercise that doesn't work.

This will help both you and us to be better.